Bank of Questions for Exams

Questions and worked solutions

25117 - Econometrics

Universitat Pompeu Fabra

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Problem 1: How dispersed are five test scores?

Five students obtain the following scores, measured in points:

$$60, 60, 70, 80, 80.$$

Treat the scores as i.i.d. observations from a population with finite variance.

1. Find the sample mean and each student's deviation from it. Check the sum of the deviations.
2. Calculate the sample variance using the divisor $n - 1$, and then the sample standard deviation. State their units.
3. What variance would the divisor n give? Explain why $n - 1$ is used when estimating the population variance, and identify the remaining degrees of freedom.

Solution 1: Mean, deviations, and sample dispersion

1. The sample mean is

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{60 + 60 + 70 + 80 + 80}{5} = \boxed{70 \text{ points}}.$$

Student i	1	2	3	4	5	Sum
$y_i - \bar{y}$	-10	-10	0	10	10	0
$(y_i - \bar{y})^2$	100	100	0	100	100	400

2. Using the sum of squared deviations,

$$s_Y^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1} = \frac{400}{5-1} = \boxed{100 \text{ points}^2}.$$

The standard deviation is the square root of the variance:

$$s_Y = \sqrt{s_Y^2} = \sqrt{100} = \boxed{10 \text{ points}}.$$

Solution 1: Why the divisor is $n-1$ (continued)

3. Dividing by n would give

$$\widetilde{s}_Y^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \overline{y})^2 = \frac{400}{5} = \boxed{80 \text{ points}^2},$$

compared with $s_Y^2 = 100$ points squared using $n-1$.

- Once the mean is fitted, $\sum_i (y_i - \overline{y}) = 0$. Knowing four of the five deviations determines the fifth: there are $n-1=4$ degrees of freedom.

- For i.i.d. observations with finite population variance σ_Y^2,

$$\mathbb{E}\left[\sum_{i=1}^{n}(Y_i - \overline{Y})^2\right] = (n-1)\sigma_Y^2.$$

  Consequently, dividing by n has expectation $\frac{n-1}{n}\sigma_Y^2$, whereas dividing by $n-1$ has expectation σ_Y^2.

Bessel's correction

The divisor $n-1$ makes the *variance estimator* unbiased. Its square root s_Y is not generally an unbiased estimator of σ_Y.